

DSS5202 Sustainable Systems Analysis**Assignment #4****Due: 19 October 2026, 9 pm****Instructions**

You may use an optimization solver to solve your model.

Copy and paste the header of its outputs, i.e., the solver name, version, model type & size, etc.

Submit your answers as one PDF file via Canvas.

Problem Description

A parcel delivery company processes three different categories of parcels:

1. General parcels
2. Fresh-food parcels
3. Fragile parcels

Currently, conventional single-use packaging is used for all three categories. Because the parcel categories have different requirements for packaging strength, thermal insulation, and impact protection, the amount of packaging material used per parcel varies across the three categories.

Table 1 shows the current packaging material usage rate for each parcel category.

Table 1: Current packaging material usage

Parcel category	Packaging material usage (kg/1000 parcels)
General parcels	360
Fresh-food parcels	640
Fragile parcels	520

To reduce the consumption of cardboard boxes, plastic bags, foam fillers, insulation materials, and other single-use packaging materials, the company is considering the following three new packaging methods that use less material:

1. **Reusable Packaging Boxes (RPB)** reduce single-use packaging consumption through collection and repeated use. This method can reduce material usage by 80%.
2. **Automated Size-Matched Packaging (ASP)** selects boxes or packaging bags according to the actual dimensions of each parcel. This method can reduce material usage by 65%.
3. **Lightweight Cushioning Materials (LCM)** reduce packaging consumption by using lighter and smaller quantities of protective fillers. This method can reduce material usage by 45%.

The costs of using the three new packaging methods vary by parcel category. Table 2 shows the packaging material and processing costs, in dollars per 1,000 parcels, for each parcel category.

Table 2: Packaging material and processing costs

Parcel category	Material and processing costs (\$/1,000 parcels)		
	RPB	ASP	LCM
General parcels	2,200	1,500	900
Fresh-food parcels	3,100	1,900	1,100
Fragile parcels	2,800	2,100	1,300

The number of parcels in each category handled annually by the company, in thousands, is given in Table 3.

Table 3: Annual volumes of parcels handled by the company

Parcel category	Annual volume handled (thousands of parcels)
General parcels	160
Fresh-food parcels	90
Fragile parcels	120

The company intends to stop using the current packaging method and has set a target to reduce its total annual packaging material consumption by at least **75%**.

It is assumed that the new packaging methods do not affect delivery operations, parcel safety, or service quality.

Question 1.

Suppose that the company may use any combination of packaging methods for each parcel category to achieve the material reduction target. Formulate an LP model to minimize the total annual cost. Show your model in standard algebraic format.

Hint: Define the decision variables: x_{ij} = annual number of category i parcels that use packaging method j . **(15 marks)**

Question 2.

Solve the LP model formulated in Question 1 using any solver. Present the optimal solution as follows: **(10 marks)**

Total annual cost: _____

The number of parcels processed annually using each method:

Parcel category	Reusable Packaging Boxes (RPB)	Automated Size-Matched Packaging (ASP)	Lightweight Cushioning Materials (LCM)
General parcels			
Fresh-food parcels			
Fragile parcels			

Current total annual packaging material (kg): _____

Annual packaging material reduced (kg): _____

Total percentage reduction achieved (%): _____

Question 3.

Suppose that each parcel category may use no more than one packaging method. Extend the LP model in Question 1 to an MIP model to incorporate this new constraint. Show your model in standard algebraic format. **(15 marks)**

Hint: Define the additional decision variables:

$$y_{ij} = \begin{cases} 1 & \text{if parcel category } i \text{ uses packaging method } j \\ 0 & \text{otherwise.} \end{cases}$$

Question 4.

Solve the MIP model formulated in Question ?? using any solver. Present the new optimal solution as follows: **(10 marks)**

Total annual cost: _____

The number of parcels processed annually using each method:

Parcel category	Reusable Packaging Boxes (RPB)	Automated Size-Matched Packaging (ASP)	Lightweight Cushioning Materials (LCM)
General parcels			
Fresh-food parcels			
Fragile parcels			

Current total annual packaging material (kg): _____

Annual packaging material reduced (kg): _____

Total percentage reduction achieved (%): _____

***** End of Assignment *****